Journal of Soils and Sediments
https://doi.org/10.1007/s11368-018-2218-6

SOILS, SEC 5 • SOIL AND LANDSCAPE ECOLOGY • RESEARCH ARTICLE

Influence of shrub roots on soil macropores using X-ray computed tomography in a shrub-encroached grassland in Northern China

Xia Hu [1,2] • Xiao-Yan Li [1,2] • Lan-Lan Guo [1,3] • Yong Liu [1,2] • Pei Wang [1,2] • Yun-Duo Zhao [1,2] • Ya-Qian Cheng [1,2] • Yan-Li Lyu [1,3] • Lian-You Liu [1,3]

Received: 22 May 2018 / Accepted: 5 December 2018

Abstract

Purpose The relationship between soil macropore and plant roots warrants special attention because it influences the behavior of water in soil. However, the influence of shrub roots on soil macropores in shrub-encroached grasslands is poorly understood. The objectives of this study were to quantify soil macropores and root architecture in a shrub-encroached grassland in northern China, and to reveal the relationship between shrub roots and soil macropore.

Materials and methods In this study, treatments were performed that corresponded to three successional states of the shrub *C. microphylla* L. with three different shrub densities. At each site, three undisturbed soil cores were excavated under the shrub canopies, and a Philips medical scanner was used to simultaneously visualize and quantify the soil and root architectures.

Results and discussion Strong positive correlations between root volume, length, and surface area and the solid surface/solid volume ratio were found, and greater root growth was noted in more porous soil. The results highlighted that the soil macropore characteristics corresponded well with the root characteristics of the soils for the three treatments. Soil macroporosity and macropore volume increased with increasing shrub root network density. In addition, the influence of plant roots on soil macropores increased with increasing shrub encroachment. The study confirmed that the large number of macropores found in the soils under shrubs was attributed to the great degree of root development there.

Conclusions The greater degree of macroporosity under shrubs was attributed to the larger root network density, which might cause greater amounts of water to be concentrated in the deep soil layer by macropore flow under shrub patches. The influence of plant roots on soil macropores increased with increasing shrub encroachment.

Keywords Computed tomography · Root architecture · Shrub encroachment · Soil macropore

1 Introduction

Soil macropores are large soil voids and pathways that consist of worm holes, root paths, soil cracks, and inter-aggregate voids, and they permit the rapid flow of water through soils (Jarvis 2007; Li et al. 2009; Helliwell et al. 2013). Soil macropores are often related to high variability in the transport of air, water, and solutes through soils. Although macropores make up only small fractions of soils (Jarvis 2007), air and water are preferentially transported through them. Soil porosity can be increased by plant roots, and preferential flow paths can be activated by plant roots, so that infiltration rates are enhanced (Feeney et al. 2006; Li et al. 2009, 2013). Many studies showed that plant roots also influence soil structure, infiltration rate, and soil nutrients (Johnson and Lehmann 2006; Li et al. 2009, 2013). Van Schaik (2009) found that soil macropores formed by the root systems of vegetation were important conduits for downward water movement in grasslands, and root channels generated soil macroporosity resulting in higher infiltration than expected based solely on the textural properties of the surface soil. Weiler and Naef (2003) reported that root channels were voids biopores formed

Responsible editor: Lu Zhang

✉ Xia Hu
huxia@bnu.edu.cn

[1] State Key Laboratory of Earth Surface Processes and Resource Ecology, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China

[2] School of Natural Resources, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China

[3] Academy of Disaster Reduction and Emergency Management, Beijing Normal University, Beijing 100875, China

Published online: 18 December 2018

by plant roots, and water flow in root channels was an important mechanism of infiltration in soil and is crucial for the prediction of runoff generation and groundwater recharge. Wu et al. (2017) indicated that, root channels resulted in macropore flow which strongly influences soil infiltration. However, until now, studies on a direct linkage between soil macropores and root architecture for the encroached *C. microphylla* L. are still lacking in the Inner Mongilia grassland.

X-ray computed tomography (CT) is a relatively new method of quantifying soil macropores (Grevers et al. 1989; Capowiez et al. 2003; Taina et al. 2008; Munkholm et al. 2012) and preferential flow (Heijs et al. 1995; Mooney and Morris 2008; Gunde et al. 2010; Sammartino et al. 2012); this technique has a higher resolution than conventional methods, such as dye tracing, spectral image analysis, and soil thin sections. Over the last decade, numerous studies related to the characterization of macropore structure (which includes properties such as macroporosity, macropore size distribution, volume, surface area, and tortuosity) have been conducted using X-ray CT for different land use and management systems (Luo et al. 2010a, b; Vogel et al. 2010; Naveed et al. 2013; Larsbo et al. 2014; Katuwal et al. 2015; Hu et al. 2015). Plant roots are sufficiently difficult to observe in situ (Clothier and Green 1997) that research has long focused on the aboveground portions of plants. Recently, researchers have attempted to observe plant roots in situ using X-ray CT. Pierret et al. (1999) examined tree roots in situ and emphasized the root architecture, but provided no specific conclusions about the structure of the root-soil interface. Mooney et al. (2012) extensively reviewed the use of X-ray CT to analyze plant roots. Kuka et al. (2013) visualized and quantified soil structure and roots simultaneously using X-ray CT. These studies suggest that X-ray CT is an appropriate method to quantify the root architecture and soil structure in undisturbed soil cores.

Shrub expansion is a common phenomenon in the grasslands of Inner Mongolia in northern China. Over 5.1×10^6 ha of grasslands are estimated to have become occupied by the shrub *C. microphylla* L. in this region (Li et al. 2013). Li et al. (2009) found that belowground shrub root channels could increase porosity and form macropore flow paths that enhance infiltration and drainage using dye-tracing method. Moreover, Li et al. (2013) suspected that the root structure of *C. microphylla* L. favors rapid water flow to deep soil layers. Hu et al. (2015) quantified the architecture of soils beneath the shrub canopy and in the adjacent interspace grass area following the establishment of a shrub *C. microphylla* L., and indicated that large numbers of soil macropores were present within shrub patches; she speculated that much soil macropores beneath the shrub canopy are mainly controlled by high root density of shrubs; however, there is no direct evidence to confirm this finding. Hu et al. (2018a) quantified the pore architecture and root architecture of grassland soils without shrub encroachment under exclosure using X-ray CT method in the Inner Mongolia grassland of northern China. Up to now, researches on the direct linkages between soil macropores and root architecture for the encroaching shrub of *C. microphylla* L. are lacking. Hence, this study aims to compare soil macropores and root architecture through applying X-ray CT to soils under encroaching *C. microphylla* L. to reveal the interactions between soil pores and root architecture in the shrub-encroached grasslands of Inner Mongolia.

2 Materials and methods

2.1 Study sites and core samples

The experiments performed in this study were conducted at the Farmland and Grassland Ecosystem Observation Station of Beijing Normal University, which is located in Taipus Banner of Inner Mongolia in northern China. This area is part of the grassland of Inner Mongolia, which is representative of the Eurasian steppe region. An analysis of 30 years of data indicates that the mean annual temperature is 1.6 °C, the maximum monthly temperature is 17.8 °C (July), and the minimum monthly temperature is − 17.6 °C (January). The annual average precipitation in this region is 407 mm, and the mean annual evaporation, as measured using a 20-cm pan, is 1750–2150 mm. According to the USDA Soil Taxonomy, the soils are Calcic-orthic Aridisols (Soil Survey Staff 1987). The dominant plants in the study area are *C. microphylla* L., *Stipa krylovii*, *Leymus chinensis*, and *Artemisia frigida* (Li et al. 2013).

Three treatments were used in the experiment. In treatment (1), the average shrub size was 0.44 ± 0.19 m^2, and shrub coverage was 1.32%; in treatment (2), the average shrub size was 1.19 ± 0.37 m^2, and shrub coverage was 12.96%; and, in treatment (3), the average shrub size was 4.34 ± 1.08 m^2, and shrub coverage was 40.12% (Peng et al. 2013; Hu et al. 2015) (Fig. 1). Sites 1, 2, and 3 represented three successional or transitional states with increasing intensity of shrub encroachment, as reflected by their increasing cover percentages of *C. microphylla* L. shrubs. In total, 9 undisturbed soil cores were taken, at the center under the canopy of *C. microphylla* L. at the three sampling sites (treatments) with three replications. The distance between sampling sites was about 200–300 m with similar terrain and soil properties. The aboveground grass and *C. microphylla* L. were clipped to ground level, and litter was completely removed to expose bare soil before soil sampling (Hu et al. 2015).

The areas of the shrub patches within each 25 m × 25 m plot were recorded on paper in the field and were then digitized using ArcGIS 9.3 to estimate the area percentages, sizes and spatial distributions of the shrub patches. The edges of the

Fig. 1 Landscapes of the field sites for the treatment 1, treatment 2, and treatment 3 (Peng et al. 2013; Hu et al. 2015)

shrub patches were visually identified in the field, based on the projected geometry of the crown shape and the sizes of *C. microphylla* L. growing in isolation or in dense clusters.

Plexiglas cylinders with a wall thickness of 4 mm, a diameter of 10 cm, and a length of 50 cm were used to obtain samples of intact soil. Core extraction followed the procedure of Hu et al. (2015, 2018a, b). The cores were extracted by pushing each cylinder slowly and vertically into the soil centimeter by centimeter. To minimize mechanical stresses during extraction, the soil around the bottom of the tube was carefully removed at each step.

The soil grain size distributions were determined using the pipette method. The organic carbon contents were determined using the method of Walkley and Black (1934). The soil characteristics identified for the three treatments are listed in Table 1.

2.2 CT scanning and image analysis

A Philips medical scanner (Brilliance ICT 256), a 128-slice spiral CT scanner, was used to scan soil cores at an energy level of 140 kV and 300 mA, which can provide detailed and low-noise projections, and 1024 × 1024 pixels per slice. The scanning mode was sequential, and the scanning spacing was 0.312 mm. After orientation, 1320 slices in the coronal view resulted from every soil column. The images were then resampled to 0.146 mm × 0.146 mm × 0.312 mm voxel cubic to facilitate computation. The Lanczos filter in Avizo version 9 (Mercury Computer Systems, Chelmsfold, MA), a widely-used resampling protocol (Meijering et al. 1999), was used to resample all the CT images. This step can improve the image quality in the depth direction by interpolating between the original images using Lanczos filter (0.625-mm slide thickness before resampling). Considering the data size and the computation ability of our computer, the cross-sectional resolution was reduced slightly from 0.312 to 0.500 mm after resampling. In addition, we found that macropore surfaces became much smoother and less jagged after resampling.

Shrub roots were visualized by a combination of the growth method and the threshold method (Hu et al. 2018b; Kuka et al. 2013). Different materials, including the acrylic cylinder, pores, roots, soil mineral particles, organic matter, microorganisms, and stones, exist within a sample. Background radiation (Illerhaus et al. 2004) and the remaining beam hardening were counted as "air or macropore", with HU (hounsfield unit) values of − 900. In the intensity histogram of raw data, three peaks could be distinguished: − 900 HU for macropores, 0 HU for shrub root, and 450 for soil material (solid) located on the root peak. The Hounsfield unit between − 650 and − 550 was obtained from the intensity histogram of the raw data by fitting a parabola, which was similar to the method of Katuwal et al. (2015). The procedure of root visualization followed the procedure of Hu et al. (2018b). An illustration of the processing steps is presented in Fig. 2.

We determined the soil macropore thresholds to obtain binary images using a manually created macropore. The determination of thresholds followed the procedure of Hu et al. (2015). A Plexiglas cylinder with a known diameter regarded as an artificial macropore was inserted into an undisturbed soil core and then pulled out. The core was then scanned using the Philips medical scanner. We selected the upper, middle, and lower parts of the Plexiglas cylinder with a known diameter respectively. First, we assumed one threshold value for the thresholding sample, then calculated the macropore size based on image analysis using Avizo 9.0 (FEI 2016), and compared it with the actual size. If the difference between the calculated and actual sizes was too large, we selected another threshold value until the difference was reduced to less than 1% (Li et al. 2013).

Slices were prepared following the procedure of Luo et al. (2010a, b) before segmentation. A median filter (radius of 3.0 pixels) was first used to remove the upper portion of the noise for all the reconstructed volumes. A cylindrical cropping tool was applied to obtain a region of interest (ROI), and then the area outside the ROI was deleted to exclude voids near the core walls and to minimize beam-hardening interference. After segmentation, the macropore networks were reconstructed and visualized using Avizo 9.0 (FEI 2016). The macropore density, macropore distribution along the column depth, macropore volume, macropore size distribution (sorted by macropore volume), mean macropore volume, and total surface area were calculated using Avizo 9.0 (FEI 2016).

Table 1 Properties of soils under shrub patches of the treatment 1 (shrub coverage, 1.32%), treatment 2 (shrub coverage, 12.96%), and treatment 3 (shrub coverage, 40.12%)

Transition stages	Average shrub size (m^2)	Shrub coverage (%)	Depth	Particle size composition (%)					Organic matter ($g\ kg^{-1}$)	Bulk density (g/cm^3)
				$0.25\ mm \le \Phi < 2.00\ mm$	$0.05\ mm \le \Phi < 0.25\ mm$	$0.02\ mm \le \Phi < 0.05\ mm$	$0.002\ mm \le \Phi < 0.02\ mm$	$\Phi \le 0.002\ mm$		
Treatment 1	0.44 ± 0.19	1.32	0–50 cm	37.92 ± 5.60	26.12 ± 1.19	9.31 ± 1.31	10.37 ± 0.55	16.28 ± 1.39	28.96 ± 1.53	1.48 ± 0.32
Treatment 2	1.19 ± 0.37	12.96	0–50 cm	17.46 ± 0.84	44.10 ± 0.83	9.90 ± 1.07	10.50 ± 1.37	16.38 ± 2.73	29.24 ± 0.96	1.34 ± 0.17
Treatment 3	4.34 ± 1.08	40.12	0–50 cm	10.76 ± 0.57	25.73 ± 1.73	16.5 ± 0.90	20.90 ± 1.80	26.10 ± 1.09	23.27 ± 1.68	1.31 ± 0.42

Average values for the different soil layers (mean ± SD)

Since soil macropores are three-dimensional (3-D), skeletonization of macropores is necessary to accurately quantify the topology and length and to effectively visualize the networks in three dimensions. The skeletons of macropore networks were generated by determining the nearest distance to the border voxels and thinning based on the nearest distance using Avizo 9 (FEI 2016).

To verify the root characteristics determined using the X-ray micro-CT images, the estimated root lengths, root surface areas, and root volumes determined from the X-ray CT images were compared with those measured by evaluation of the roots in the soil cores. The roots were extracted from the nine soil cores by disaggregating the cores and manually washing them with a 250-mm sieve to separate out the individual roots. The roots were then scanned and analyzed using WinRHIZO 2009 (Regent Instruments) to quantify specific parameters of the roots (specifically, their lengths, areas and volumes) (Arsenault et al. 1995).

2.3 Statistical analyses

All statistical analyses were conducted using SPSS software (version 13.0, SPSS Inc., Chicago, IL). Differences among the measured parameters between the different treatments were analyzed using Fisher's protected least significant difference (LSD) test. The formula for the least significant difference is:

$$LSD_{A,B} = t_{0.05/2,DFW}\sqrt{MSW\left(1/n_A + 1/n_B\right)} \quad (1)$$

where t is critical value from the t-distribution table, MSW is mean square within, obtained from the results of the ANOVA test, and n is the number of scores used to calculate the means.

3 Results

3.1 Visualization of root networks and root density

The relationships among the root lengths, root surface areas, and root volumes estimated using X-ray CT and the values of these parameters obtained using the standard root washing method are shown in Fig. 3. Good relationships existed between the two methods with respect to root length, for which the calculated coefficient of determination (R^2) was 0.723; similar results were obtained for root surface area, for which $R^2 = 0.708$, and for root volume, for which $R^2 = 0.924$. The results showed that X-ray CT is a suitable method for performing root evaluations in situ, due to its relatively fine resolution of the roots.

Three-dimensional visualizations of the roots in the nine soil cores are shown in Fig. 4. The roots under the shrubs were large and long and extended to great depths. In addition, large numbers of roots were distributed in deep soil layers under the

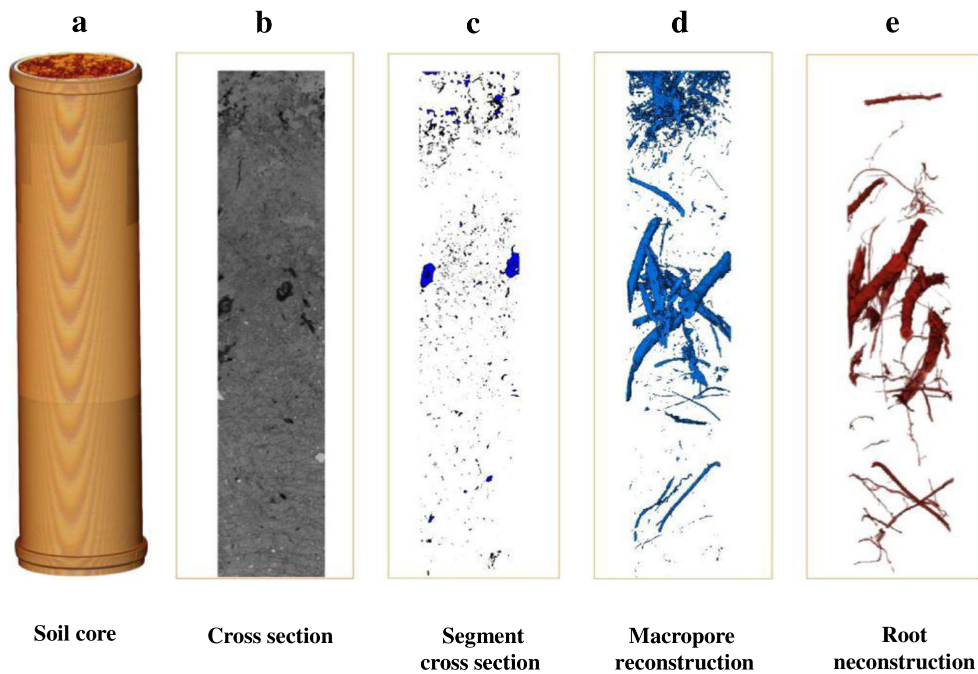

Fig. 2 Steps in image processing: **a** scanned soil monolith (50 cm in height and 10 cm in diameter); **b** two-dimensional cross-section showing the internal structure as a greyscale image; **c** two-dimensional cross-section after segmentation, in which black and white indicate pores and the soil matrix, including solids and unresolved pore space, respectively; **d** the reconstructed pore network; and **e** the reconstructed root network (Hu et al. 2019)

shrubs. Moreover, the roots were also large and long and extended to great depths under treatment 3 than under treatments 1 and 2. The plant roots in the soil profiles under the shrub canopy tended to be complex, vertically oriented, and continuous in the soil profiles.

The root density differed significantly among the three degrees of shrub encroachment (Table 2). The soil under the *C. microphylla* L. shrubs under treatment 1 displayed a root density of 0.00741 $mm^3\ mm^{-3}$; the corresponding values for treatments 2 and 3 were 0.01013 $mm^3\ mm^{-3}$ and 0.02040 $mm^3\ mm^{-3}$, respectively. The root surface area and root network density varied among the three degrees of shrub encroachment according to a pattern that resembled that of root density (Table 2). The shrub patches under treatment 3 displayed the greatest root surface area ($6.0468 \times 10^4\ mm^2$). Similarly, the soil under the *C. microphylla* L. shrubs under treatment 1 displayed a network density of 0.0473×10^5 number mm^{-3}; the corresponding values for treatments 2 and 3 were 0.0751×10^5 number mm^{-3} and 0.1387×10^5 number mm^{-3}, respectively. The root length density and root node density were calculated from the skeletonized roots (Table 2). As expected, the mean root length density and node density were highest for the shrub patches under treatment 3, with values of 3.124×10^3 km m^{-3} and 4.6042×10^5 number m^{-3}, respectively (Table 2). Significant differences in root density were found among the three treatments. There were no significant differences in total surface, network density and length density among the three treatments.

No significant differences in mean volume root size were observed among the three replicates of each treatment examined in this study (Table 2). The soils under the *C. microphylla* L. shrubs under treatment 1 displayed a mean volume root size of $2.1597 \times 10^4\ mm^3$; corresponding values of $1.8666 \times 10^4\ mm^3$ and $6.1989 \times 10^4\ mm^3$ were obtained for treatments 2 and 3, respectively. The soil under the *C. microphylla* L. shrubs under treatment 1 displayed a mean root equivalent diameter of 9.682 mm, and the corresponding values for treatments 2 and 3 were 7.149 mm and 8.200 mm, respectively.

To investigate the variation of root density with depth, root density distribution for each sample was calculated slice per slice, i.e., for every 0.312-mm depth interval. The distributions of root density with sample height (profile depth) for

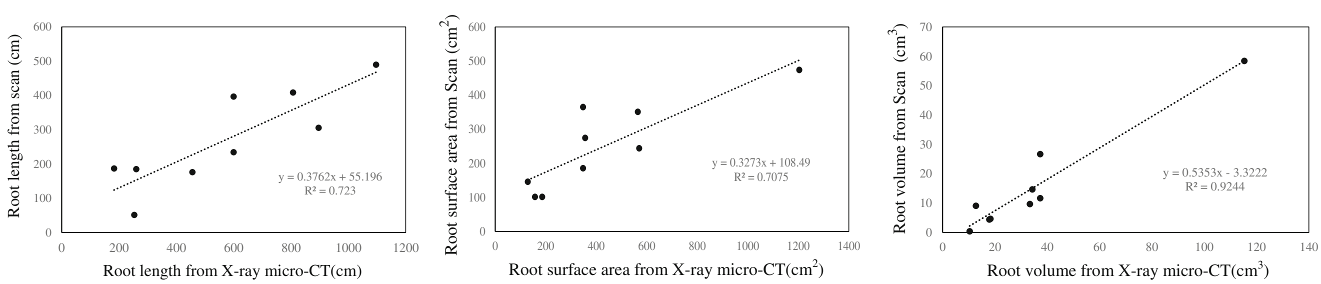

Fig. 3 Comparison of the root length, root surface area, and root volume from X-ray CT and from scanning after washing

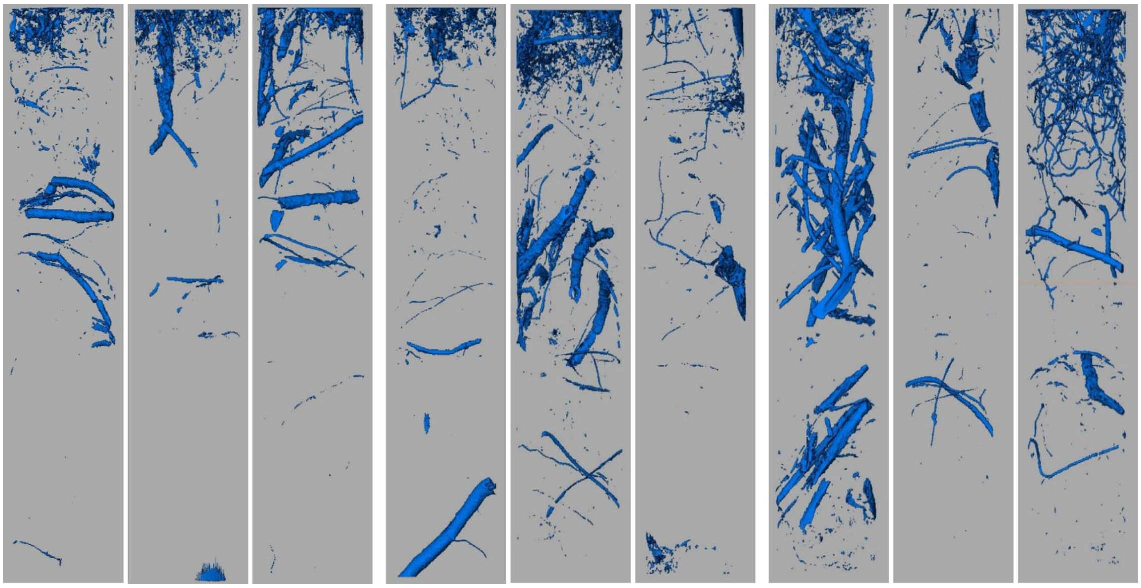

Fig. 4 Three-dimensional visualization of root networks in the soil cores for shrub patches in the treatment 1 (shrub coverage, 1.32%), treatment 2 (shrub coverage, 12.96%), and treatment 3 (shrub coverage, 40.12%). Blue indicates roots, whereas gray represents non-pore regions

soils of the three treatments are presented in Fig. 5. For the soils under the shrub canopy under treatments 1 and 2, the root density declines sharply between soil depths of 0–70 mm, increases between soil depths of 70–250 mm, and is slightly undulatory as the soil depth increases further. The root density increases over soil depths ranging from 0 to 250 mm and then increases over a range of soil depths of 250–400 mm; it remains stable at soil depths ranging from 400 mm to 450 mm (treatment 3). In general, the data showed that root density displayed higher variability for soils under the shrub canopy of treatment 3, due to the dense root systems of shrubs.

3.2 Proportions of macropores and roots in undisturbed soil cores

Table 3 presents data on soil macropore volume, soil macropore surface area, root volume, and root surface area determined from the X-ray CT scans. The proportions of soil macropores are in the range of 0.4285–1.3522%, and the plant root volumes are in the range of 0.7410–2.0401% (Table 3). No significant differences were noted for the surface areas of solids or the volumes of roots among all of the samples.

The soils under the shrub patches under treatment 3 displayed the highest root surface area (mean of 19.3116 $m^2\ m^{-3}$) together with a high root volume (mean 0.0204 $m^3\ m^{-3}$) (Table 3). There were no significant differences in root surface area among all of the plots (t test, $P < 0.05$) (Table 3). The soils under the shrub patches under the treatment 3 displayed the highest macropore surface areas (mean of 10.1112 m^2 root surface m^{-3} soil) and high macropore volumes (mean 1.3522 m^3 root volume m^{-3} soil) (Table 3).

3.3 Relationship between roots and soil macropores

Three-dimensional visualizations of soil macropores and plant roots are shown in Fig. 6. The distributions of roots in the soil profiles were consistent with the soil macropores. Larger

Table 2 Root characteristics of soil cores determined from X-ray CT images for shrub patches of the treatment 1 (shrub coverage, 1.32%), treatment 2 (shrub coverage, 12.96%), and treatment 3 (shrub coverage, 40.12%)

Transition stages	Root density ($mm^{-3}.mm^{-3}$)	Total surface area ($\times 10^4 mm^2$)	Mean root equivalent diameter (mm)	Network density ($\times 10^5$ number.mm^{-3})	Length density ($\times 10^3$ km.m^{-3})	Mean root size ($\times 10^4$ mm^3)	Node density ($\times 10^5$ number m^{-3})
Treatment 1	0.00741 (± 0.00241) c	2.1424 (± 0.7139)	9.682 (± 0.438)	0.0473 (± 0.0116)	1.4402 (± 0.7152)	2.1597 (± 0.6514)	1.8704 (± 0.0651)
Treatment 2	0.010133 (± 0.00464) b	2.5409 (± 0.8989)	7.149 (± 2.385)	0.0751 (± 0.0252)	1.2339 (± 0.4820)	1.8666 (± 0.6227)	2.1677 (± 0.0623)
Treatment 3	0.02040 (± 0.00948) a	6.0468 (± 1.5923)	8.200 (± 1.637)	0.1387 (± 0.0706)	3.1244 (± 0.6740)	6.1989 (± 2.6698)	4.6042 (± 0.2670)

Columns for each data group followed by the same letter are not significantly different at $P < 0.05$, and the absence of a letter indicates no significant difference for that property. Values in parentheses represent standard deviations

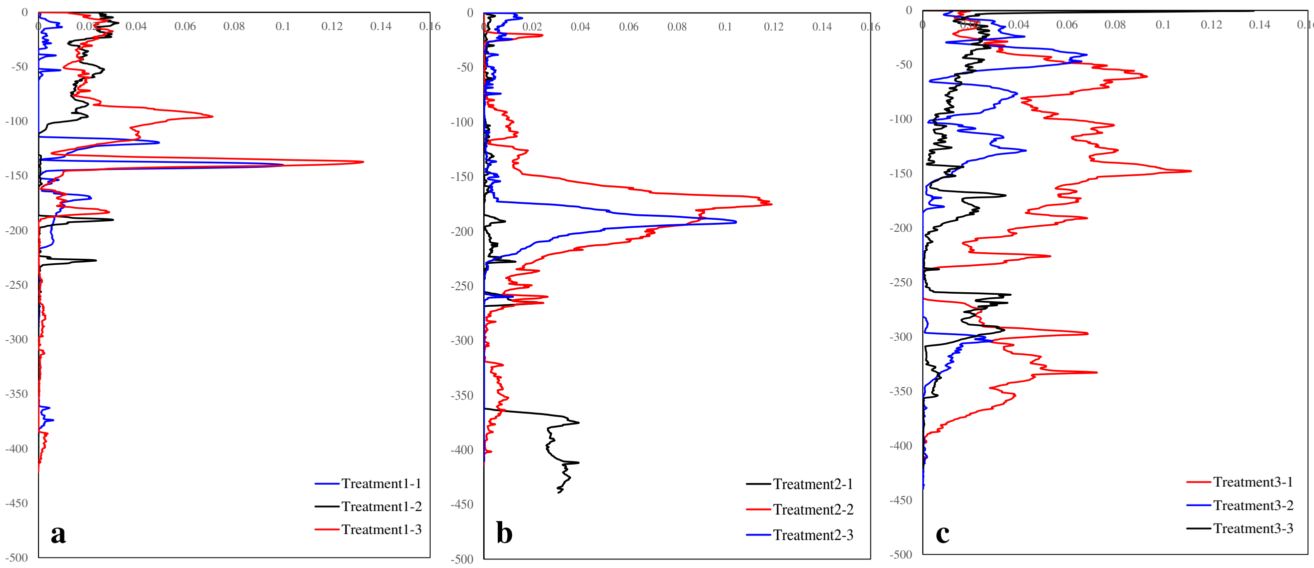

Fig. 5 Distribution of CT-measured root density and SD as a function of depth within the soil column for shrub patches in the treatment 1 (shrub coverage, 1.32%), treatment 2 (shrub coverage, 12.96%), and treatment 3 (shrub coverage, 40.12%)

numbers of macropores and roots were concentrated in the deep soil layers under the shrub patches. The roots and soil macropores under the shrub canopy were present at depths of 50–400 mm. More soil macropores and roots were long and extended to great depths. The roots and soil macropores in the soils under the *C. microphylla* L. shrubs were more abundant under treatment 3 than under treatments 1 and 2. Therefore, the spatial distribution of the roots was consistent with that of the soil macropores.

The results of correlation analyses between the soil macropore parameters and the root parameters in the soils under the shrubs are shown in Table 4. For the soils under the shrubs, the macroporosity, hydraulic radius, and mean macropore volume were significantly and positively correlated with the root network density (Table 4). In addition, the macropore number and macropore network density were significantly and positively correlated with the number of roots. And, the macropore number were significantly and negatively correlated with root size. Thus, the roots had a great effect on the soil macropores in the soils under the shrub patches.

4 Discussion

X-ray CT scanning provides 3D visualizations of plant roots, i.e., root volume and area (Young 1998; Carminati et al. 2009; Tracy et al. 2010, 2012; Schmidt et al. 2012). The findings of this study indicated that the spatial distributions of the roots were consistent with those of the soil macropores. The results also demonstrated close relationships between the soil macropores and plant root volume and root surface area; moreover, the characteristics of soil structure were roughly in accordance with the characteristics of the roots under the three treatments. During sampling, the phenomenon that roots preferentially grow close to macropores in soil matrix, in direct contact with macropore walls (Fig. 7), was observed in situ, consistent with previous reports (Stewart et al. 1999).

The correlations between soil macroporosity and root density within the soils under the shrub canopies are shown in Fig. 8. The parameter of root density was significantly correlated ($R^2 = 0.49$) with the estimated soil macroporosity in the soils under the shrub canopies under treatment 1; the corresponding values under

Table 3 Results of X-ray computed microtomography image analysis and comparisons with the root washing and scanning method

Transition stages	Surface		Pores % (v/v)	Root volume		Root volume % (v/v)	Root surface	
	Root surface $m^2\ m^{-3}$	Macropore surface $m^2\ m^{-3}$		CT m^3 root m^{-3}	Scanning		CT m^2 root m^{-3}	Scanning soil volume
Treatment 1	7.2571	7.3809	0.4285	0.0074	0.0028	0.7410	7.2571	6.1022
Treatment 2	8.9972	4.1999	0.7516	0.0065	0.0023	0.6502	8.9972	6.8460
Treatment 3	19.3116	10.1112	1.3522	0.0204	0.0326	2.0401	19.3116	9.9200

Columns for each data group by the same letter (lowercase and uppercase for columns and rows, respectively) are not significantly different at $p = 0.05$; the absence of letters indicates no significant differences for that property. Data in the parentheses are standard deviations

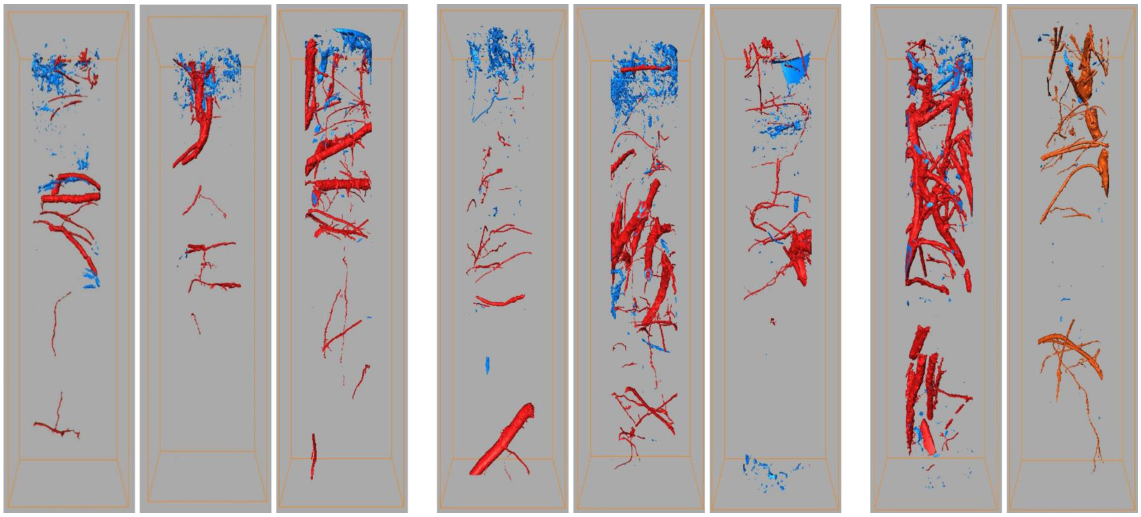

Fig. 6 Three-dimensional visualization of root and macropore networks in soil cores for shrub patches for the treatment 1 (shrub coverage, 1.32%), treatment 2 (shrub coverage, 12.96%), and treatment 3 (shrub coverage, 40.12%)

treatments 2 and 3 were 0.58 and 0.65, respectively (Fig. 8). Therefore, as the intensity of shrub encroachment increased, the influence of plant roots on soil macropores also increased. These findings support our hypothesis that the growth of the roots of the shrub *C. microphylla* L. lead to the development of large numbers of macropores in deep soil layers (Li et al. 2013).

Plant roots had a great effect on soil macropores within the shrub patches (Table 4). The small number of replicates (three replicates) in the study perhaps could affect the characteristics of soil macropores, but the result is consistent with the report of Hu et al. (2015), who noted that the soils under shrub patches displayed larger numbers of soil macropores, which would contribute to greater root development. Under the shrub canopies, the soil macroporosity and mean macropore volume were significantly and positively correlated with the root network density. Under the shrub canopies, the soil macroporosity and macropore volume increased with increases in the root network density of the shrubs. The finding

Table 4 Correlation coefficients between macropore and root properties under shrub patches

Root parameters	Soil macropore parameters										
	Macroporosity	Total surface area	Network density	Macro size	Length density	Node density	Hydraulic radius	Macropore volume	Total length	Node number	Macropore number
Root density	0.917	0.893	−0.139	0.374	0.720	0.359	0.922	0.940	0.810	0.524	−0.005
Root surface area	0.967	0.815	0.011	0.231	0.816	0.495	0.970	0.980	0.889	0.646	0.146
Root network density	0.999*	0.678	0.218	0.025	0.918	0.664	0.999*	1.000**	0.964	0.790	0.347
Root size	−0.435	0.415	−0.984	0.912	−0.722	−0.947	−0.423	−0.380	−0.618	−0.872	−0.999*
Root length density	0.900	0.911	−0.180	0.413	0.690	0.320	0.905	0.924	0.785	0.488	−0.046
Root node density	0.968	0.811	0.018	0.224	0.820	0.501	0.971	0.982	0.892	0.651	0.152
Root label volume	0.916	0.894	−0.142	0.377	0.718	0.356	0.921	0.939	0.809	0.522	−0.007
Root total length	0.956	0.838	−0.029	0.270	0.792	0.460	0.959	0.972	0.870	0.615	0.106
Root node number	0.958	0.834	−0.022	0.263	0.796	0.466	0.961	0.973	0.873	0.621	0.113
Number of roots	0.335	−0.510	0.997*	−0.951	0.644	0.907	0.323	0.278	0.530	0.814	0.998*

*Correlation is significant at the 0.05 level

**Correlation is significant at the 0.01 level

Fig. 7 Soil profile under shrub canopy

of this study also suggests that we can infer soil macropore structure under the *C. microphylla* L. canopy by knowing root density using CT method and/or sampling system.

Under treatment 1, the soils under the *C. microphylla* L. shrubs displayed a mean root size of 2.1597×10^4 mm^3; the corresponding values for treatments 2 and 3 were 1.8666×10^4 mm^3 and 6.1989×10^4 mm^3, respectively. Moreover, different root types have different functions (Pierret et al. 2007; Zobel et al. 2007). Large-diameter roots are typically comparatively long and few in number, and they are the main pathways by which water and nutrients are transported to shoots (Blouin et al. 2007). In contrast, small-diameter roots are typically comparatively short and numerous and play major roles in taking up water and nutrients (Yanai and Eissenstat 2002; Blouin et al. 2007). Therefore, the large numbers of macropores found in the soils under the shrubs can be attributed to greater degrees of root development, and we speculated that roots are the main pathways by which water and nutrients are transported. Moreover, shrub patches can trap sediments carried by overland flow and wind, resulting in higher clay contents in the soils under shrub canopies. This process may have a direct effect on microbial communities because soils with higher clay contents can preserve or protect microbial biomass (Gallardo and Schlesinger 1992).

The node density varied among the soil type-land use combinations following a pattern that was similar to those of the macroporosity and the macropore mean angle. Higher macroporosity values are associated with larger node density values (Table 2), indicating correspondingly higher degrees of inter-connectivity among the macropores. Among all of the soil columns studied, the soils of the shrub patches displayed relatively continuous macropore networks. This result is consistent with the report of Peng et al. (2013) in the same study site, who performed a dye tracing experiment showing that shrub patches favor the development of preferential water infiltration, which permits water to be transported faster to deeper soil layers, where it is stored. In addition, Peng (2012) reported that the average rainfall interception percentage at the study site is 21% for *C. microphylla* L. shrubs and 7.5% for grass patches. These results are consistent with

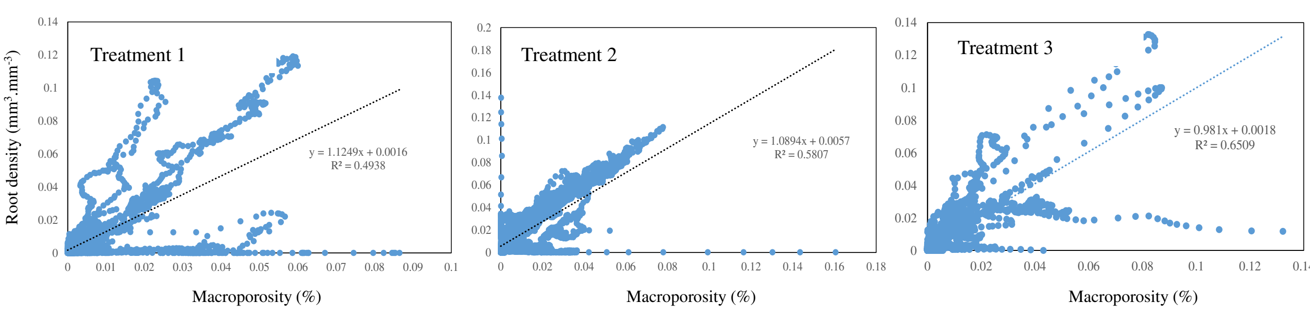

Fig. 8 Relationship between soil macroporosity and root density under shrub canopies in the treatment 1 (shrub coverage, 1.32%), treatment 2 (shrub coverage, 12.96%), and treatment 3 (shrub coverage, 40.12%)

previous findings by Dunkerley (2000), who reported that the water uptake rate within shrub patches is at least 10% greater than between the shrubs.

5 Conclusions

The aim of this study was to use X-ray CT to quantify the influence of shrub roots on soil macropores in the shrub-encroached grasslands of Inner Mongolia in northern China. X-ray computed microtomography was used to visualize and quantify the soil structure and roots architecture in areas displaying three degrees of shrub encroachment. The micromorphological images showed that the distributions of roots were consistent with the distributions of soil macropores. The results also demonstrated that the soil macropore parameters roughly corresponded to the root parameter characteristics of the soils under the three treatments used in this study. The effects of roots on the soil macropores were great under the shrub canopies. In addition, the influence of plant roots on soil macropores increased with increasing degrees of shrub encroachment. The macroporosity was significantly and positively correlated with the root network density in soils under the shrub canopies. The large numbers of macropores found in the soils under the shrubs can be attributed to the greater degree of root development there.

Funding information This study was financially supported by National Key R&D Program of China (Grant number: 2016YFA0601901), the National Science Foundation of China (Grant number: 41471018), the PCSIRT (IRT-15R06), the National Key Research and Development Program of China (2016YFE0203400-01), and projects supported by the State Key Laboratory of Earth Surface Processes and Resource Ecology.

References

Arsenault JL, Poulcur S, Messier C, Guay R (1995) WinRHIZO™, a root measuring system with a unique overlap correction method. HortScience 30:906

Blouin M, Barot S, Roumet C (2007) A quick method to determine root biomass distribution in diameter classes. Plant Soil 290:371–381

Capowiez Y, Pierret A, Moran CJ (2003) Characterisation of the three-dimensional structure of earthworm burrow systems using image analysis and mathematical morphology. Biol Fertil Soils 38:301–310

Carminati A, Vetterlein D, Weller U, Vogel HJ, Oswald SE (2009) When roots lose contact. Vadose Zone J 8:805–809

Clothier BE, Green SR (1997) Roots: the big movers of water and chemical in soil. Soil Sci 162(8):534–543

Dunkerley D (2000) Hydrologic effects of dryland shrubs: defining the spatial extent of modified soil water uptake rates at an Australian desert site. J Arid Environ 45:159–172

Feeney DS, Crawford JW, Daniell T, Hallett PD, Numan N, Ritz K, Rivers M, Young IM (2006) Three-dimensional microorganization of the soil–root–microbe system. Microb Ecol 52(1):151–158

FEI (2016) Amira & Avizo 3D software. [WWW Document]. URL. https://www.fei.com/software/amira-avizo/. Accessed date: 1 March 2017

Gallardo A, Schlesinger WH (1992) Carbon and nitrogen limitations of soil microbial biomass in desert ecosystems. Biogeochemistry 18:1–17

Grevers MCJ, Jong ED, St Arnaud RJ (1989) The characterization of soil macroporosity with CT scanning. Can J Soil Sci 69(3):629–637

Gunde AC, Bera B, Mitra SK (2010) Investigation of water and CO_2 (carbon dioxide) flooding using micro-CT (micro-computed tomography) images of Berea sandstone core using finite element simulations. Energy 35:5209–5216

Heijs AWJ, de Lange J, Schoute JFT, Bouma J (1995) Computed tomography as a tool for non-destructive analysis of flow patterns in macroporous clay soils. Geoderma 64:183–196

Helliwell JR, Sturrock CJ, Grayling KM, Tracy SR, Flavel RJ, Young IM, Whalley WR, Mooney SJ (2013) Applications of X-ray computed tomography for examining biophysical interactions and structural development in soil systems: a review. Eur J Soil Sci 64:279–297

Hu X, Li ZC, Li XY, Liu Y (2015) Influence of shrub encroachment on CT-measured soil macropore characteristics in the Inner Mongolia grassland of northern China. Soil Tillage Res 150:1–9

Hu X, Li ZC, Li XY, Wang P, Zhao YD, Liu LY, Lv YL (2018a) Soil macropore structure under different land uses characterized by x-ray computed tomography in the Qinghai Lake watershed, NE Qinghai-Tibet plateau. Pedosphere 28(3):478–487

Hu X, Lyu YL, Liu Y, Li XY, Li ZC, Sun ZT, Cheng YQ, Guo LL, Liu LY (2018b) Exclosure on CT-measured soil macropore characteristics in the Inner Mongolia grassland of northern China. J Soils Sediments 18:718–726

Hu X, Li X Y, Wang P, Liu Y, Wu X C, Li Z C, Zhao Y D, Cheng Y Q, Guo L L, Lyu Y L, Liu L Y (2019) Influence of exclosure on CT-measured soil macropores and root architecture in a shrub-encroached grassland in northern China. Soil & Tillage Research 187:21–30

Illerhaus B, Onel Y, Goebbels A (2004) Correction techniques for 2D detectors to be used with high energy X-ray sources for CT, part II. In: Developments in X-ray tomography proceedings of the society of photo-optical instrumentation engineers (SPIE) 5535:329–334

Jarvis NJ (2007) A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. Eur J Soil Sci 58(3):523–546

Johnson MS, Lehmann J (2006) Double-funneling of trees: Stemflow and root-induced preferential flow. Ecoscience 13(3):324–333

Katuwal S, Norgaard T, Moldrup P, Lamandé M, Wildenschild D, de Jonge LW (2015) Linking air and water transport in intact soils to macropore characteristics inferred from X-ray computed tomography. Geoderma 237–238:9–20

Kuka K, Illerhaus B, Fox CA, Joschko M (2013) X-ray computed microtomography for the study of the soil-root relationship in grassland soils. Vadose Zone J 12. https://doi.org/10.2136/vzj2013.01.0014

Larsbo M, Koestel J, Jarvis N (2014) Relations between macropore network characteristics and the degree of preferential solute transport. Hydrol Earth Syst Sci 18:5255–5269

Li XY, Yang ZP, Li YT, Lin H (2009) Connecting ecohydrology and hydropedology in desert shrubs: Stemflow as a source of preferential flow in soils. Hydrol Earth Syst Sci 13:2023–2030

Li XY, Hu X, Zhang ZH, Peng HY, Zhang SY, Li GY, Li L, Ma YJ (2013) Shrub hydropedology: preferential water availability to deep soil layer. Vadose Zone J 12(4):1539–1663

Luo L, Lin H, Li S (2010a) Quantification of 3D soil macropore networks in different soil types and land uses using computed tomography. J Hydrol 393:53–64

Luo L, Lin H, Schmidt J (2010b) Quantitative relationships between soil macropore characteristics and preferential flow and transport. Soil Sci Soc Am J 74:1929–1937

Meijering EHW, Niessen WJ, Pluim JPW, Viergever MA (1999) Quantitative comparison of sinc-approximating kernels for medical image interpolation. In: Medical Image Computing and Computer-Assisted Intervention, MICCAI'99, Proceedings, vol. 1679:210–217

Mooney SJ, Morris C (2008) Morphological approach to understanding preferential flow using image analysis with dye tracers and X-ray computed tomography. Catena 73(2):204–211

Mooney SJ, Pridmore TP, Helliwell J, Bennett MJ (2012) Developing X-ray computed tomography to non-invasively image 3-D root systems architecture in soil. Plant Soil 352:1–22

Munkholm LJ, Heck RJ, Deen B (2012) Soil pore characteristics assessed from X-ray micro- CT derived images and correlations to soil friability. Geoderma 181–182:22–29

Naveed M, Moldrup P, Emmanuel A, Wildenschild D, Edene M, Lamandé M, Vogel HJ, de Jonge LW (2013) Revealing soil structure and functional macroporosity along a clay gradient using x-ray computed tomography. Soil Sci Soc Am J 77:403–411

Peng HY (2012) Spatial pattern of shrub patches and its ecohydrological mechanism at the typical steppe in Inner Mongolia. Unpublished PhD thesis. Beijing Normal University, China, pp 47–48

Peng HY, Li XY, Li GY, Zhang H, Zhang S Y, Li L, Zhao GQ, Jiang ZY, Ma YJ (2013) Shrub encroachment with increasing anthropogenic disturbance in the semiarid Inner Mongolian grasslands of China. Catena 109: 39–48

Pierret A, Capowiez Y, Moran CJ, Kretzschmar A (1999) X-ray computed tomography to quantify tree rooting spatial distributions. Geoderma 90:307–326

Pierret A, Doussan C, Capowiez Y, Bastardie F, Pages L (2007) Root functional architecture: a framework for modeling the interplay between roots and soil. Vadose Zone J 6:269–281

Sammartino S, Michel E, Capowiez Y (2012) A novel method to visualize and characterize preferential flow in undisturbed soil cores by using multi-slice helical CT. Vadose Zone J 11(1):vzj2011.0100

Schmidt S, Bengough AG, Gregory PJ, Grinev DV, Otten W (2012) Estimating root–soil contact from 3D X-ray microtomographs. Eur J Soil Sci 63:776–786

Soil Survey Staff (1987) Keys to soil taxonomy, 3rd edn. Cornell University, Ithaca

Stewart JB, Moran CJ, Wood JT (1999) Macropore sheat: quantification of plant root and soil macropore association. Plant Soil 211:59–67

Taina IA, Heck RJ, Elliot TR (2008) Application of X-ray computed tomography to soil science: a literature review. Can J Soil Sci 88(1):1–20

Tracy SR, Roberts JA, Black CR, McNeill A, Davidson R, Mooney SJ (2010) The X-factor: visualizing undisturbed root architecture in soils using X-ray computed tomography. J Exp Bot 61:311–313

Tracy SR, Black CR, Roberts JA, Sturrock C, Mairhofer S, Craigon J, Mooney SJ (2012) Quantifying the impact of soil compaction on root system architecture in tomato (*Solanum lycopersicum L.*) by X-ray micro-computed tomography (CT). Ann Bot 110:511–519

Van Schaik NLMB (2009) Spatial variability of infiltration patterns related to site characteristics in a semi-arid watershed. Catena 78:36–47

Vogel MN, Goeppert B, Maksimovic O, Brodoefel H, Faul C, Claussen CD, Horger M (2010) CT features of neutropenic enterocolitis in adult patients with hematological diseases undergoing chemotherapy. Fortschr Röntgenstr 182(12):1076–1081

Walkley A, Black IA (1934) An examination of the Degtjareff method for determining soil organic matter and a proposed modification of the chromic acid titration method. Soil Sci 37(1):29–38

Weiler M, Naef F (2003) An experimental tracer study of the role of macropores in infiltration in grassland soils. Hydrol Process 17(2): 477–493

Wu GL, Liu Y, Yang Z, Cui Z, Deng L, Chang XF, Shi ZH (2017) Root channels to indicate the increase in soil matrix water infiltration capacity of arid reclaimed mine soils. J Hydrol 546:133–139

Yanai RD, Eissenstat DM (2002) Coping with herbivores and pathogens: a model of optimal root turnover. Funct Ecol 16(6):865–869

Young IM (1998) Biophysical interactions at the root-soil interface: a review. J Agric Sci 130:1–7

Zobel RW, Baligar VC, Kinraide TB (2007) Fine root diameters can change in response to changes in nutrient concentrations. Plant Soil 297:243–254